

Task 6: Password Strength Evaluation

ElevateLabs Cybersecurity Internship

1. Objective

The objective of this task is to understand what makes a password strong by creating passwords of varying complexity and evaluating them using an online password strength checker (passwordmeter.com). The results are used to identify best practices for creating secure passwords and to understand how password complexity affects resistance to common attacks.

2. Tool Used

Password Strength Checker: passwordmeter.com

This free online tool evaluates password strength based on length, character variety, patterns, and common usage. It provides a rating (Very Weak to Very Strong), entropy in bits, and estimated time to crack under different attack scenarios.

3. Passwords Tested and Results

Six passwords were created with varying complexity and tested on passwordmeter.com. Results are summarised in the table below.

#	Password	Rating	Entropy	Tool Feedback	Crack Time (Fast DB)
1	123456	Very Weak	~undef	Most commonly used password	Instantly
2	P@ssw0rd!	Very Weak	~29 bits	Keyboard pattern; under 12 chars	9 hours
3	Pass@1234	Fair	~66 bits	Under 12 chars; all 4 char types	6 years
4	Summer#2024	Weak	~64 bits	Contains date/year; easily guessed	2 years
5	BlueSky!Rain#42	Strong	~98 bits	Good length; no common patterns	45B years
6	Tr0ub4dor&3#Secure!	Very Strong	~125 bits	Excellent length; all char types	>1 trillion yrs

4. Key Observations

- **Common passwords crack instantly.** '123456' is in every attacker's dictionary — it was flagged as one of the most commonly used passwords and cracked in zero time under all scenarios.
- **Predictable patterns are dangerous.** 'P@ssw0rd!' looks complex but uses a keyboard walk pattern. Despite having all four character types, it rated Very Weak due to the pattern and short length.

- **Years and seasons add little security.** 'Summer#2024' was flagged for containing a recognisable date pattern, dropping it to Weak despite using all character types.
- **Length dramatically increases strength.** Moving from 9 characters (P@ssw0rd!) to 15 (BlueSky!Rain#42) jumped the rating from Very Weak to Strong and cracked-time from hours to billions of years.
- **The strongest password had no patterns.** 'Tr0ub4dor&3#Secure!' scored ~125 bits of entropy and would take over 1 trillion years to crack under any scenario.

5. Common Password Attacks

Brute Force Attack

The attacker tries every possible combination of characters systematically. Short passwords (under 8 chars) can be brute-forced in seconds with modern hardware. Each extra character multiplies the search space exponentially — a major reason length matters.

Dictionary Attack

The attacker uses a pre-built list of common words, names, and known passwords. Passwords like '123456', 'password', 'Summer2024', and 'monkey' are all in standard dictionaries. Simple substitutions (@ for a, 0 for o) are also covered in modern dictionaries.

Credential Stuffing

Attackers use username/password pairs leaked from previous data breaches and try them on other sites. This is why reusing passwords across accounts is dangerous even if the password itself is strong.

Rainbow Table Attack

A precomputed table of hash values is used to reverse-engineer passwords from stolen hash databases. Weak encryption (MD5/SHA-1) combined with a common password can be cracked instantly — as seen with '123456' in the test results.

6. Best Practices for Strong Passwords

- Use at least **12–16 characters** — length is the single most impactful factor.
- Include all four character types: **uppercase, lowercase, numbers, and symbols.**
- Avoid **dictionary words, names, keyboard patterns** (qwerty, 123456), and dates.
- Use **passphrases** — a string of 4+ random words is long, memorable, and strong.
- Never **reuse passwords** across accounts. A breach on one site exposes all others.
- Use a **password manager** (e.g., Bitwarden, KeePass) to generate and store unique passwords.
- Enable **Multi-Factor Authentication (MFA)** wherever possible as a second layer of defence.
- Avoid **personal information** — birthdays, pet names, and phone numbers are easily guessed.

7. Summary

This task demonstrated clearly that password strength is not just about including symbols — it is about **length, unpredictability, and avoiding known patterns**. A password like 'P@ssw0rd!' may feel secure but is trivially cracked. In contrast, a long, random password like 'Tr0ub4dor&3#Secure!' provides over 1 trillion years of protection even against fast hardware. The most practical takeaway is to use a password manager that generates truly random, long passwords, combined with MFA for all critical accounts.